

# Cardiovascular Risk Prediction Using Machine Learning

Farhan Sadik

*Department of Computer Science*

*BRAC University*

*Dhaka, Bangladesh*

ID: 24101406

farhan.sadik2@g.bracu.ac.bd

MD Tohsin Mridha

*Department of Computer Science*

*BRAC University*

*Dhaka, Bangladesh*

ID: 23201572

mohammad.tohsin.mridha@g.bracu.ac.bd

Md Tariful Islam Talha

*Department of Computer Science*

*BRAC University*

*Dhaka, Bangladesh*

ID: 22201334

tariful.islam.talha@g.bracu.ac.bd

**Abstract**—The prediction of cardiovascular risk assists in determining individuals who are vulnerable to developing severe cardiovascular issues. This project's objective is to demonstrate the capability of machine learning in assessing cardiovascular risk. The dataset contains 1,529 records belonging to individuals with 22 features. Cardiovascular risk is divided into three levels: LOW, INTERMEDIARY, and HIGH. In the data preprocessing steps, missing information, both quantitative and qualitative data is analyzed, meaningful information regarding the health condition of individuals is obtained, and factors that may lead to distortion and misinterpretation of results are removed. Four models were chosen for the analysis: Logistic Regression, Random Forest, AdaBoost, and Neural networks. Comparison of the results obtained using those models with K nearest neighbor and Gradient boosting methods is made after the implementation of PCA for dimensionality reduction. A parallel regression model is also generated to predict the CVD Risk Score. According to the recent tests, the Neural Network has the highest accuracy level of about 67.97%. Then, the second place goes to Logistic Regression with approximately 65.36% correct predictions, while the technique showing the lowest classification performance is AdaBoost at 64.38%. In the case of Random Forest, applying PCA changes its performance from 58.82% to 62.42%, while the classifiers mentioned above lower their accuracy scores. The performed analysis shows that classifiers have “issues” with the identification of the LOW category, which has a significantly smaller representation than other classes. The regression method, however, reaches the  $R^2$  value of 0.951. The results have shown that accuracy alone is insufficient for evaluating model performance, particularly under class imbalance.

**Index Terms**—Cardiovascular Risk Prediction, Machine Learning, Logistic Regression, Random Forest, AdaBoost, Neural Network, Principal Component Analysis, Multiclass Classification

## I. INTRODUCTION

Cardiovascular diseases (CVDs) are diseases affecting the heart and the blood vessels and are still a major concern in medicine. Most cardiovascular problems are related to identifiable risk factors such as age, blood pressure, cholesterol level, diabetes, smoking, BMI, physical activity level, and family history. Identifying people with different levels of cardiovascular risk can be helpful in taking timely preventive measures and in improving the quality of clinical decision-making. Traditional risk evaluation tools, such as QRISK2 and the AHA PREVENT equations, use a combination of known

clinical factors to evaluate the level of cardiovascular risk [1], [2].

Machine learning offers another way to evaluate cardiovascular risk. Machine learning allows assessing risk by learning patterns and relationships from patient records. In contrast to traditional approaches that use a particular statistical formula, machine learning algorithms have an advantage of being able to use different variables in their predictions. Various machine learning algorithms have been used for risk prediction purposes, including Logistic Regression, Random Forest, Support Vector Machine, AdaBoost, Neural Network, XGBoost, and ensemble algorithms [3]–[5].

Recently, research has been conducted on improving predictions of cardiovascular risk using machine learning algorithms in Bangladesh. In a comparative study done by Islam et al., findings have shown that XGBoost provided the best results when identifying individuals with high cardiovascular risk [10]. In another study, Tasnim et al. conducted cardiovascular risk assessments on data from 1,529 patients by applying a stacking model that combined traditional methods with explainable AI [12]. Based on above-mentioned studies, it is implied that machine learning technologies can serve as great tools for cardiovascular risk assessment but are still to be used in practice.

The current research investigates the identification of cardiovascular risk as a multiclass classification task consisting of three classes: LOW, INTERMEDIARY, and HIGH risk. The proposed approach relies on a dataset including structured clinical and demographic data from a total of 1,529 individuals. Several types of models such as Logistic Regression, Random Forest, AdaBoost, and Neural Network are utilized to evaluate the difference between classification models in terms of their ability to assess cardiovascular risk levels effectively.

## II. LITERATURE REVIEW

Cardiovascular risk prediction has been studied through traditional statistical models, regular machine learning techniques, deep learning, and ensemble methods. Earlier studies offer helpful information about the important cardiovascular risk factors, choice of model, risk categorization into different

classes, evaluation methods, and constraints to consider in machine learning models.

Hippisley-Cox developed QRISK2 with the help of Cox proportional hazards regression based on the data of almost 2.3 million people and revealed ROC numbers of around 0.817 for females and about 0.792 for males [1]. The model is important for our research as some predictors of QRISK2 such as age, smoking, blood pressure, cholesterol, BMI, and diabetes are common risk factors of cardiovascular diseases. Nevertheless, unlike LOW, INTERMEDIARY, and HIGH in our study, QRISK2 estimates risk of prospective events using longitudinal survival model.

Khan et al. created the PREVENT formulas of the American Heart Association, using information from over six million individuals, and found C-statistics of about 0.794 for women and 0.757 for men for total CVD [2]. The article shows the significance of common risk factors for CVD that can be obtained from the clinical tests and gives a solid conventional risk reference for this study. On the other hand, PREVENT is a kind of risk equation based on survival rather than a multielement classifier.

Unnikrishnan et al. compared the Framingham method to locally trained linear models and Support Vector Machines in the assessment of the risk for CVD and determined that SVM was more efficient in terms of sensitivity and specificity [3]. The study matters for the project in that it utilized traditional clinical parameters, such as cholesterol level, blood pressure, BMI, smoking, diabetes, and hypertension, for the machine learning risk analysis. It conducted the research with a local cohort and did not check a more advanced multi-target cardiovascular risk model.

A study by Cho et al., using 222,998 Korean adults, compared the performance of Logistic Regression, TreeBag, Random Forest, AdaBoost, and Neural Networks and found Neural Network was the best performer with a C-statistic of 0.751 [4]. The relevance of this study arises from the fact that many algorithms from the study are also utilized in our project and the scope of the project indirectly measures the performance of models in cardiovascular prevention. The limitation of the report is that it is conducted using target population of statin naive Korean adults and hence the results cannot be mixed with study data of other populations.

A study by Alaa et al. places the AutoPrognosis algorithm in testing with the 423,604 participants of UK Biobank and achieved the measure of AUC-ROC of 0.774 as compared to 0.724 from the Framingham score [5]. The work provides evidence that machine learning can use traditional and non-traditional predictors for improvement of cardiovascular risk monitoring behavior. The limitation of the research is that hundreds of candidate variables have been tested, which makes the research much more complicated than our regular testing.

In a study performed by Li et al. involving about 3.6 million participants, the results showed comparable population-level C-statistics of about 0.87 despite significant variations in personal risk forecasts [6]. This implies that while selecting an appropriate model requires more than a mere assessment

of universal performance metrics; several other crucial factors must be used to make predictions. Although this research focused mainly on long-term event prediction with censoring, the current study is concerned more with distinguishing cardiovascular risk categories.

While Barbieri et al. integrated deep learning and survival analysis to predict five-year cardiovascular risk for over 2.1 million adults in New Zealand, it turned out that the algorithms based on deep learning performed better than standard Cox models in such aspects as explained variance, calibration, and discrimination [7]. The study indicates that the use of more flexible learning algorithms can have its benefits, but there are some downsides related to the database used for it that faced limitations as it lacked some direct clinical measurements such as those related to blood pressure and lipid levels that are available in the structured database we possess.

Neumann and colleagues used Cox regression with LASSO variable selection to create a cardiovascular risk model for healthy adults over 70 years of age and validated the development with an AUC of 0.681 and with 0.642 for external validation [8]. This study is notable because it pinpoints critical cardiovascular risk factors such as age, smoking, systolic blood pressure, cholesterol, creatinine, and diabetes and emphasizes the need for external validation in the model validations. However, the limitation is that it focuses on a particular age group, which makes it less applicable when it comes to datasets that involve bigger ages.

Suri and his colleagues conducted a review study examining 265 articles that surveyed different multiclass, multilevel, and ensemble machine learning technologies for cardiovascular risk assessment and found that feature extraction methods and ensemble methods are being used widely [9]. The significance of this review is linked to the fact that our classification involves three classes of cardiovascular risk and falls into the same multiclass risk assessment type discussed in the paper. However, it is worth noting that Suri's work is a review paper and has no specific dataset and results to share.

Islam et al. utilized BDHS 2022 information collected from a group of 2,221 subjects in Bangladesh to compare Logistic Regression, Naive Bayes, ANN, Random Forest, and XGBoost, with the latter returning an accuracy of 68.22% and an AUC score of 0.721 [10]. The article is highly relevant because it is directed solely at cardiovascular risk prediction in Bangladesh, as well as featuring some feature selection and balancing techniques before comparing models. However, its high-risk results only stem from hypertension and diabetes data and will not allow for the longitudinal assessment of risk because of the data being cross-sectional.

Shah et al. presented a hybrid ensemble model applying Gradient Boosting, CatBoost, LightGBM, SVM, Neural Networks, and XGBoost as a meta-learner, with the results providing an AUC-ROC equal to 0.82 [11]. The significance of this study lies in its demonstration of the potential of ensemble learning and SHAP-based explainability in enhancing not only cardiovascular risk prediction but also interpretation of the important risk factors involved. The multi-model approach

and numerous datasets used create much greater complexity compared to the baseline models used in the current project.

Tasnim et al. have created a stacking regression model by integrating Ridge regression, Theil Sen regression, and Gradient boosting methods using a sample of 1,529 patients from Bangladesh [12]. They achieved an R-squared value of about 0.96 after conducting cross-validation. The findings are significant since the design of the data set and categorization of patients according to low, intermediate, and high risk is similar to that of our study. The authors, however, adopted a regression approach to evaluate the risk categorization, and employed data from a single health institution making way for the future research work to explore the same risk aspect in a multiclassing classification perspective.

### III. DATASET AND PREPROCESSING

#### A. Dataset Description

This dataset includes clinical, lifestyle, and demographic data from 1,529 subjects gathered in 22 features on physical information (age, height, weight, BMI, abdominal girth), clinical data (blood pressure, cholesterol, HDL, blood sugar, LDL), and life/history data (smoking, diabetes, exercise, CVD family history). Each record bears two potential outputs: categorical CVD Risk Level (LOW, INTERMEDIARY, HIGH) applying in the main classification task, and continuous CVD Risk Score applying in another regression task. The information in the table indicates that there is a considerable class imbalance, since only 14.4% of the records belong to LOW-risk group.

TABLE I  
CVD RISK LEVEL CLASS DISTRIBUTION

| Risk Level   | Count | Percentage |
|--------------|-------|------------|
| HIGH         | 728   | 47.6%      |
| INTERMEDIARY | 581   | 38.0%      |
| LOW          | 220   | 14.4%      |
| Total        | 1,529 | 100%       |

#### B. Data Cleaning, Feature Recovery, and Preprocessing

The first inspection indicated that there was no repetition of records, however, some fields had a moderate amount of data missing (in particular, the report showed that 82 values of Diastolic BP were missing, 81 of the weight, and 80 of HDL). Instead of throwing away any values the data analyst restored the missing data, therefore, blood pressure values were extracted from a combined column, height in meters was calculated based on the height in centimeters without data being altered, and BMI values were computed by putting the height and weight together. Then, the column with the blood pressure, the duplicating height and both risk indicators were removed to avoid loss of valuable data, thus, 18 possible variables were left (12 of them numerical and 6 categorical).

The results of the exploratory analysis revealed a tendency towards a normal distribution for almost all numerical variables, with the exception of BMI and CVD Risk Score distributions. The two variables had distributions that were closer to normal. The gender distribution was found to be approximately balanced (773 females and 756 males). Our

correlation analysis suggested that the variables of height in meters and centimeters are highly redundant ( $r = 1.00$ ). Total Cholesterol and Estimated LDL have a correlation of 0.96 and the correlation of Abdominal Circumference and Waist-to-Height Ratio is  $r = 0.90$ . On the other hand, CVD Risk Score has the strongest linear links with variables like BMI, Total Cholesterol, and Systolic BP among others.

The feature set was cleaned and divided into the training data that consisted of 80% of the data (1,223 records) and the testing data consisting of 20% of the data (306 records). Upon acquiring the data the ColumnTransformer pipeline developed in scikit-learn was applied. The ColumnTransformer was trained only on the training data and median imputed and standardized numerical features and was also one-hot-encoded categorical features.

### IV. METHODOLOGY

#### A. Logistic Regression

A multinomial Logistic Regression classifier (scikit-learn, `max_iter = 1000`) was fit on the preprocessed features as an interpretable linear baseline.

#### B. Random Forest

A Random Forest was implemented from scratch on top of `DecisionTreeClassifier` base learners to control the bagging and feature-randomization logic explicitly. Each of 25 trees (max depth 5) is fit on a bootstrap sample using a random subset of  $\sqrt{n}$  features per split; predictions are decided by majority vote.

#### C. AdaBoost (Implemented from Scratch)

A multiclass AdaBoost classifier based on the SAMME algorithm was implemented from scratch using decision stumps with `max_depth = 1` as weak learners. AdaBoost operates sequentially, with each weak learner receiving a training distribution that gives greater emphasis to observations misclassified by previous learners.

For a training set containing  $N$  observations, every observation initially receives the same weight:

$$w_i^{(1)} = \frac{1}{N}, \quad i = 1, 2, \dots, N. \quad (1)$$

At boosting iteration  $t$ , the weighted error of the decision stump  $h_t$  is calculated as

$$\epsilon_t = \frac{\sum_{i=1}^N w_i^{(t)} \mathbb{I}(y_i \neq h_t(x_i))}{\sum_{i=1}^N w_i^{(t)}}. \quad (2)$$

For the  $K = 3$  cardiovascular risk classes, the SAMME amount-of-say assigned to the accepted weak learner is

$$\alpha_t = \ln \left( \frac{1 - \epsilon_t}{\epsilon_t} \right) + \ln(K - 1). \quad (3)$$

The sample weights are then updated according to whether each training instance was classified correctly:

$$w_i^{(t+1)} = \frac{w_i^{(t)} \exp[\alpha_t \mathbb{I}(y_i \neq h_t(x_i))]}{Z_t}. \quad (4)$$

where the normalization factor  $Z_t$  is

$$Z_t = \sum_{j=1}^N w_j^{(t)} \exp[\alpha_t \mathbb{I}(y_j \neq h_t(x_j))]. \quad (5)$$

Thus, incorrectly classified observations receive relatively larger weights and become more influential when the next decision stump is fitted.

After the boosting rounds are complete, classification is performed using the weighted vote of the accepted weak learners:

$$\hat{y}(x) = \arg \max_{k \in \{1, \dots, K\}} \sum_{t=1}^T \alpha_t \mathbb{I}(h_t(x) = k). \quad (6)$$

The implementation uses a maximum of 50 boosting rounds. Weak learners whose weighted errors are at or above the multiclass chance threshold are rejected, while the accepted stumps contribute to the final prediction according to their  $\alpha_t$  values.

#### D. Neural Network

The network includes a Dense layer containing 64 units using the ReLU activation function, followed by Dropout with a rate of 0.2, a subsequent Dense layer of 32 units using ReLU activation, followed by Dropout with a 0.2 dropout rate. Finally, the third Dense layer consists of 16 ReLU units. The last layer has three softmax outputs representing the three classes of risks: HIGH, INTERMEDIARY, and LOW.

The modeling problem is solved using Adam, with a learning rate of 0.001 and the sparse categorical cross-entropy as a loss function. We have 20% of the original dataset left as test set, and the training data has been separated into two parts: training and validation. The model is trained in at most 300 epochs with a mini-batch size of 32. EarlyStopping method was configured with patience=15 and restored the best weights.

#### E. Dimensionality Reduction with PCA

PCA is employed on the selected numeric features instead of the entire one-hot-encoded feature set. The pre-processing process produces 12 numeric features as well as 15 categorical encoded features. PCA is trained on the numerical training set and is set up to maintain at least 95% of the numerical variance. As a result of such a transformation, 9 principal components are retained having 96.25% of variance from the original 12 numeric components. After that, these 9 features are combined with the unchanged 15 categorical encoded features, which leads to a PCA-based representation with 24 components. The classifiers of the original models are trained again using this new representation, which enables the dimensionality reduction to be evaluated without transforming

categorical variables instead of projecting them in the PCA space.

#### F. Additional Models on PCA-Reduced Features

Two further scikit-learn classifiers were trained using the PCA-based representation: K-Nearest Neighbors ( $k = 15$ , distance-weighted) and Gradient Boosting (100 estimators, learning rate 0.1, max depth 3).

#### G. Regression Model for CVD Risk Score

A Random Forest Regressor (200 estimators, max depth 8) was trained inside the same pipeline to predict the continuous CVD Risk margin directly, using the records with a valid score (80/20 split: 1,167 training, 292 testing), isolating the difficulty introduced by discretizing the score into three risk levels.

### V. EXPERIMENTAL SETUP

#### A. Training Configuration

All classifiers apply the 80/20 stratified arrangement (with `random_state` set to 42). The Neural Network additionally incorporates a validation data set as well as early stopping to ensure convergence and to prevent over-fitting. All hyperparameters of the classifiers are presented in Table II.

TABLE II  
HYPERPARAMETER CONFIGURATION FOR EACH MODEL

| Model                   | Key Hyperparameters                                                            |
|-------------------------|--------------------------------------------------------------------------------|
| Logistic Regression     | <code>max_iter = 1000</code>                                                   |
| Random Forest (scratch) | 25 trees, max depth 5, $\sqrt{n}$ feature sampling                             |
| AdaBoost (scratch)      | 50 stumps, depth 1, SAMME                                                      |
| Neural Network          | 64–32–16 ReLU, dropout 0.2, Adam ( $lr = 0.001$ ), max 300 epochs, patience 15 |
| KNN                     | $k = 15$ , distance-weighted                                                   |
| Gradient Boosting       | 100 estimators, $lr = 0.1$ , max depth 3                                       |
| RF Regressor            | 200 estimators, max depth 8                                                    |

#### B. Evaluation Metrics

We use several performance metrics to evaluate classification results. In addition to accuracy, we present precision, recall, and F1-score. Accuracy alone might not be satisfactory, due to class imbalance discussed in Table I. Therefore, we also rely on confusion matrices to determine the most confusing classes. ROC curves for one class vs rest classification are employed to determine class difference at various thresholds. Regression performance is measured using three performance indicators: MAE, RMSE, and ( $R^2$ ).

### VI. RESULTS AND ANALYSIS

#### A. Overall Performance Comparison

Table III depicts the latest accuracy data regarding the four baseline models. From the data presented, it may be seen that the Neural Network performs the best with 67.97% accuracy; Logistic Regression takes the second place at 65.36% accuracy, while AdaBoost and Random Forest come third and fourth with 64.38% and 58.82% respectively.

TABLE III  
OVERALL TEST ACCURACY OF THE FOUR BASELINE MODELS

| Model               | Test Accuracy |
|---------------------|---------------|
| Logistic Regression | 65.36%        |
| Random Forest       | 58.82%        |
| AdaBoost            | 64.38%        |
| Neural Network      | 67.97%        |

### B. Class-wise Performance Analysis

The precision and recall according to the risk class are given in Table IV. The performance for each class with the Logistic Regression, Random Forest, and AdaBoost is the same as in the previous study. On the contrary, the new Neural Network yields considerable advances. It reaches HIGH risk recall of around 0.77, INTERMEDIARY recall of 0.72, LOW risk precision of almost 0.52 and recall of 0.25, and LOW risk precision of about 0.25. Although the LOW risk remains the hardest one, the modified Neural Network has shown much greater performance.

TABLE IV  
CLASS-WISE PRECISION (P) AND RECALL (R)

| Model    | HIGH |      | INTERMEDIARY |      | LOW  |      |
|----------|------|------|--------------|------|------|------|
|          | P    | R    | P            | R    | P    | R    |
| LR       | 0.71 | 0.75 | 0.63         | 0.74 | 0.31 | 0.09 |
| RF       | 0.61 | 0.74 | 0.56         | 0.62 | 0.00 | 0.00 |
| AdaBoost | 0.66 | 0.77 | 0.63         | 0.73 | 0.00 | 0.00 |
| NN       | 0.71 | 0.77 | 0.67         | 0.72 | 0.52 | 0.25 |

The results from the confusion matrices presented in Fig. 1 confirm the results represented in Table IV. The Random Forest and AdaBoost fail again to predict LOW risk cases, but Logistic Regression finds very few of them. The modern Neural Network provides a more balanced confusion matrix and has improved LOW risk detection without losing HIGH and INTERMEDIARY performance. However, the majority of LOW risk cases incorrectly classified as INTERMEDIARY, still confirming the idea that LOW/INTERMEDIARY boundary is the greatest obstacle in classification.

### C. Effect of Dimensionality Reduction (PCA)

The revised PCA technique only uses numerical features. Nine principal components account for about 96.25% of the variance among the original 12 numerical features, whereas the 15 one-hot-encoded categorical features remain unchanged. This results in a PCA-based feature representation of 24 dimensions. Table V shows the results of the four baseline models before and after the modification. Random Forest achieves the largest improvement: from 58.82% to 62.42%. Logistic Regression drops from 65.36% to 64.05%, AdaBoost - from 64.38% to 59.48%, and the Neural Network - from 67.97% to 65.69%. Therefore, PCA improves the effectiveness of Random Forest but does not contribute to improving the performance of the other three baseline models. Table VI compares the results of the most successful version of each of the six algorithms evaluated. The neural network without PCA takes first place with the score equal to 67.97%, followed by Logistic Regression with 65.36% and AdaBoost with 64.38%. After PCA, Random Forest shows the best performance among the four models at the level of 62.42%. Gradient Boosting and KNN, both evaluated on the PCA-based representation, achieve 61.11% and 58.17%, respectively.

TABLE V  
EFFECT OF PCA ON TEST ACCURACY

| Model               | Before PCA | After PCA | $\Delta$ (pts) |
|---------------------|------------|-----------|----------------|
| Logistic Regression | 65.36%     | 64.05%    | -1.31          |
| Random Forest       | 58.82%     | 62.42%    | +3.59          |
| AdaBoost            | 64.38%     | 59.48%    | -4.90          |
| Neural Network      | 67.97%     | 65.69%    | -2.29          |

TABLE VI  
OVERALL COMPARISON ACROSS ALL SIX EVALUATED MODELS

| Model                   | Test Accuracy |
|-------------------------|---------------|
| Neural Network          | 67.97%        |
| Logistic Regression     | 65.36%        |
| AdaBoost                | 64.38%        |
| Random Forest (PCA)     | 62.42%        |
| Gradient Boosting (PCA) | 61.11%        |
| KNN (PCA)               | 58.17%        |

### D. ROC Curve Analysis

ROC analysis examines all the model expressions. Assessment of models, Logistic Regression, Random Forest, AdaBoost, and Neural Network has been made both with PCA and without PCA, while KNN and Gradient Boosting have been evaluated using PCA representation. The results are presented in the form of ROC graphs and per-class AUC data, while Table VII represents average AUC data for the models. In accordance with the results, the updated Neural Network reaches the highest average AUC of 0.81 without using PCA, and the Neural Network with PCA is the second with 0.78. Random Forest scores 0.75 without PCA and 0.74 with PCA. Logistic regression scores 0.73 without PCA and 0.72 with PCA, while the rest of the experiments yield average AUC values of around 0.69-0.72. The ROC results give an additional explanation regarding classification behavior. Specifically, the improved Neural Network can successfully classify LOW-risk instances much better than LOW recall could let expect. The difference between final AUC and multiclass recall shows that there is useful information about probabilities, though the default decision rule might not always choose LOW as the final classification.

TABLE VII  
MACRO-AVERAGE ROC AUC BY MODEL

| Model / Configuration            | Macro-Avg AUC |
|----------------------------------|---------------|
| Neural Network — Before PCA      | 0.81          |
| Neural Network — After PCA       | 0.78          |
| Random Forest — Before PCA       | 0.75          |
| Random Forest — After PCA        | 0.74          |
| Logistic Regression — Before PCA | 0.73          |
| Logistic Regression — After PCA  | 0.72          |
| KNN — After PCA                  | 0.72          |
| Gradient Boosting — After PCA    | 0.72          |
| AdaBoost — Before PCA            | 0.70          |
| AdaBoost — After PCA             | 0.69          |

### E. CVD Risk Score Regression

Table VIII displays how well the Random Forest Regressor predicted the continuous CVD Risk Score. The newly

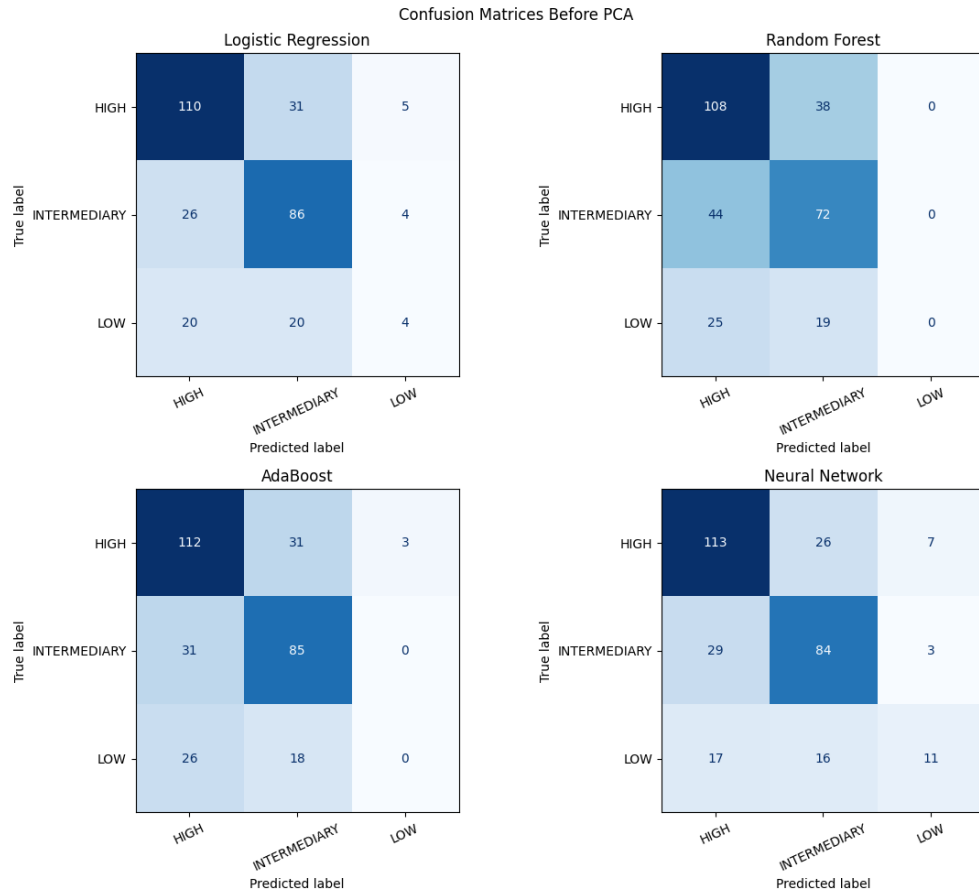


Fig. 1. Confusion matrices of the four baseline models (before PCA) on the test set.

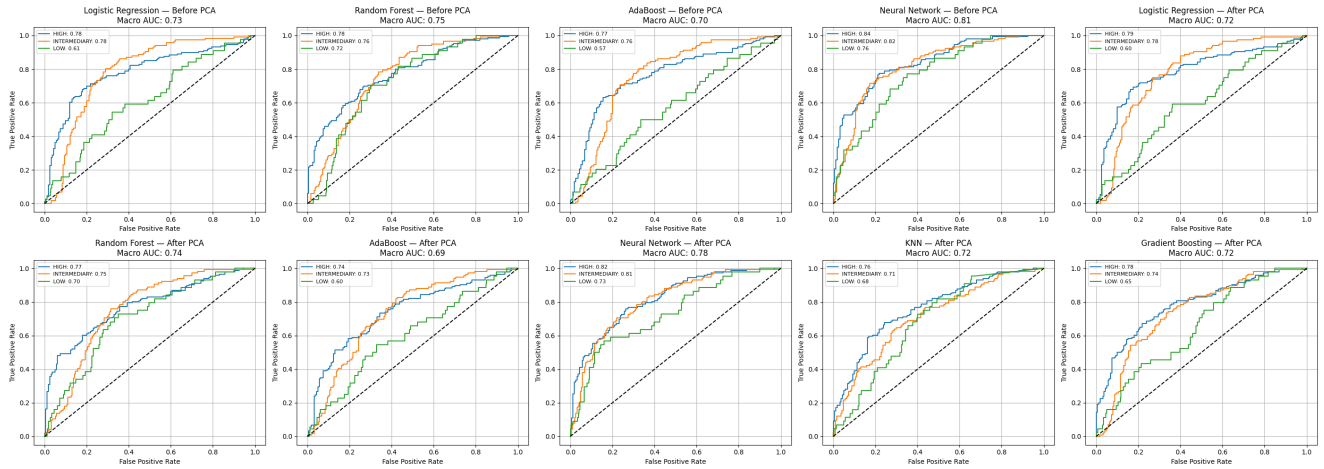


Fig. 2. One-vs-rest ROC curves and per-class AUC for the evaluated model/PCA configurations.

created notebook has also continued to perform really well:  $R^2 = 0.951$ ,  $MAE = 0.408$ ,  $RMSE = 0.544$ . Figure 3 shows the predicted and actual scores where most of the points are relatively close to the ideal diagonal line.

This finding is in line with the  $R^2 = 0.96$  calculated by Tasnim et al. based on the same data of 1,529 patients [12]. The high performance of the continuous score in comparison with the multi-class classification performance indicates that

the main difficulty is associated with classifying the cardiovascular risk score into LOW, INTERMEDIARY, and HIGH parameters.

TABLE VIII  
REGRESSION PERFORMANCE FOR CVD RISK SCORE

| Metric | Value |
|--------|-------|
| MAE    | 0.408 |
| RMSE   | 0.544 |
| $R^2$  | 0.951 |

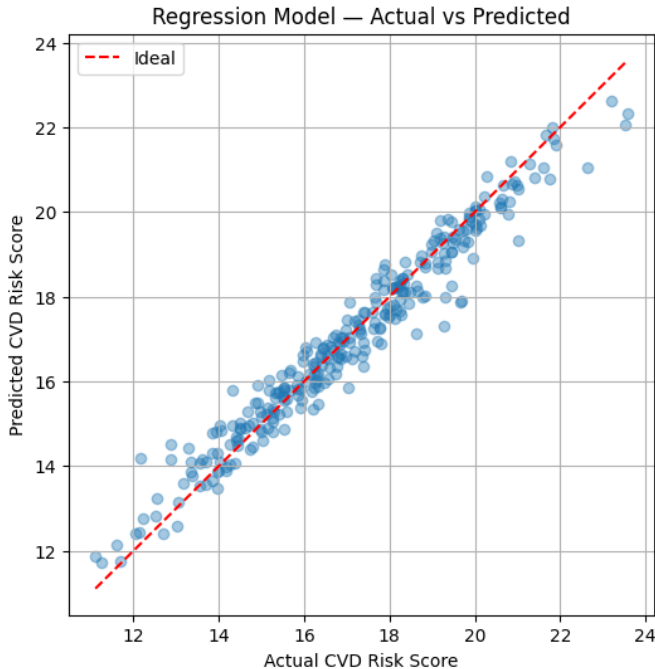


Fig. 3. Regression model—actual vs. predicted CVD Risk Score on the test set.

## VII. DISCUSSION

### A. The LOW-Risk Class Remains the Main Bottleneck

The LOW-risk category is the hardest class among all classifiers to get correctly classified by the classification algorithm. It accounts for just 14.4% of the overall dataset and has 44 instances in the testing dataset. Random Forest and AdaBoost did not classify LOW-risk classes correctly before using PCA, while Logistic regression has a LOW-class recall of only 9%.

Neural Network improves upon this, having a LOW-class precision of around 0.52 and recall of approximately 0.25. Although this is still significantly lower than the accuracy achieved for other categories like HIGH and INTERMEDIATE risk classes, this shows that using deeper architecture with dropout and early stopping can substantially enhance the performance for the smallest class. The new analysis run on Colab also shows that accuracy is not the only relevant metric for HIGH-risk patients.

Random Forest performs PCA and achieves a HIGH-risk recall of around 78.08%, which is associated with a 21.92 % false negative rate. On the contrary, the Neural Network with no PCA scores HIGH-risk recall of approximately 77.40%. Thus, even though the model has the highest accuracy, it does not have the lowest number of false negative cases with clinical significance.

### B. Why the Updated Neural Network Performed Best

The revised Neural Network achieves the highest overall test accuracy at 67.97%, replacing Logistic Regression as the strongest classifier in the updated experiments. The deeper 64–32–16 architecture gives the model greater capacity to represent nonlinear relationships among the clinical and lifestyle variables, while the two Dropout layers and early stopping provide explicit mechanisms for reducing overfitting.

The Neural Network also obtains the highest macroaverage ROC AUC in the latest testing, reaching almost 0.81 before PCA. Hence, its benefit is not only limited to the precise final classification but also allows it to successfully separate the three risk levels based on its probability measure.

### C. Overfitting Control in the Neural Network

Neural Network tend to diverge increasingly in the training and validation accuracies. After introducing two Dropout layers and an EarlyStopping callback that can limit training to a maximum of 300 epochs, but stop the process when validation loss is not decreasing for 15 epochs, and performs weight restoration based on the best validation performance. Those changes bring about a substantial improvement and increased the test accuracy to 67.97%. Nevertheless, it is still important to monitor validation as this larger architecture can easily overfit the small dataset without regularization.

### D. Limited and Model-Dependent Benefit of Dimensionality Reduction

PCA technique is better suited to the mixed tabular dataset because it changes only numerical attributes while maintaining the one-hot-encoded categorical features. Nine numerical principal components account for 96.25% of the numerical variability, and along with the 15 relevant categorical features, it provides a 24-dimensional input representation. The result is model-specific.

The Random Forest model is the greatest benefactor of the preprocessing technique since it exhibits a significant improvement from 58.82% to 62.42%. Unlike the previous case, Logistic Regression, the AdaBoost, and Neural Network models do not improve after PCA. Thus, the subsequent data transformations eliminate some valuable information for a few classifiers while providing Random Forest with a more compact representation of numerical data.

### E. Comparison with Related Work

Our latest classification accuracy stands at 67.97%, nearly equal to the 68.22% accuracy reported by Islam et al. [10] for a related cardiovascular-risk dataset in Bangladesh. Nevertheless, this comparison must be approached with caution since the datasets, targets, features, and evaluation methods differ. The regression results are equally impressive. In particular, the Random Forest Regressor achieves an  $R^2$  value near 0.951, which is in the vicinity of the 0.96 performance recorded by Tasnim et al. [12]. This substantial discrepancy between the performance of continuous variable regression and three-class classification shows that the main difficulty in classification

only begins once the continuous risk score has been converted into LOW, INTERMEDIARY, and HIGH categories.

## VIII. CONCLUSION AND FUTURE WORK

### A. Conclusion

In this study, machine learning techniques implemented on 1,529 patient records were compared for cardiovascular risk assessment. The results of the latest study show that a revised Neural Network has the best testing accuracy (67.97%), while Logistic Regression and AdaBoost follow suit with 65.36% and 64.38%, respectively.

The method of PCA that uses numbers only gives mixed outcomes. The Random Forest method yields an improvement from 58.82% to 62.42%, but the other three methods worked better on their own. The results of the KNN and Gradient Boosting methods in PCA give 61.11% and 58.17%, respectively.

The Neural Network has the highest macro-average ROC AUC for the group of models considered with a macro average AUC of 0.81 when no PCA is used, as well as the best LOW-risk performance in classification compared to other models tested. However, LOW-risk is still a drawback category.

The recent research shows that clinical errors should be evaluated separately from the total accuracy. For instance, Random Forest gives even better results regarding HIGH risk evaluation as compared to the leading Neural Network.

Ultimately, the Random Forest Regression model gets an  $R^2$  value of 0.951 in predicting CVD Risk Scores. This remarkable outcome shows that the relation of cardiovascular risk can be modeled much better than the three-way discretized classification task.

### B. Future Work

Future attempts may try to resolve the issue of class imbalance in a direct fashion through the oversampling techniques such as SMOTE, usage of class-weighted loss functions, or adjusting the decision thresholds guided by ROC curves for each class. The thresholding technique may also be an effective way of decreasing the number of false negatives in the HIGH-risk group.

Utilizing more optimized algorithms to substitute both Random Forest and AdaBoost as well as employing gradient-boosting methods like XGBoost and LightGBM may help in closing the remaining gap in accuracy to some extent. Considering the competent regression capability, a hybrid approach may be considered involving a primary predictive step to identify the continuous CVD Risk Score and thereby identifying the category of risk using defined thresholds. Treating the three risk categories of the CVD (LOW, INTERMEDIARY and HIGH) as ordinal rather than purely categorical might better explain the interrelation between these three categories.

Last but not least, additional tuning of the neural network hyperparameters as well as external validation on an independent sample cohort would help determine whether it is possible to further improve results on a different dataset.

## REFERENCES

- [1] J. Hippisley-Cox, C. Coupland, Y. Vinogradova, J. Robson, R. Minhas, A. Sheikh, and P. Brindle, "Predicting cardiovascular risk in England and Wales: prospective derivation and validation of QRISK2," *BMJ*, vol. 336, no. 7659, pp. 1475–1482, 2008, doi: 10.1136/bmj.39609.449676.25.
- [2] S. S. Khan et al., "Development and validation of the American Heart Association's PREVENT equations," *Circulation*, vol. 149, pp. 430–449, 2024, doi: 10.1161/CIRCULATIONAHA.123.067626.
- [3] P. Unnikrishnan, D. K. Kumar, S. Poosapadi Arjunan, H. Kumar, P. Mitchell, and R. Kawasaki, "Development of health parameter model for risk prediction of CVD using SVM," *Computational and Mathematical Methods in Medicine*, vol. 2016, Art. no. 3016245, 2016, doi: 10.1155/2016/3016245.
- [4] S.-Y. Cho et al., "Pre-existing and machine learning-based models for cardiovascular risk prediction," *Scientific Reports*, vol. 11, Art. no. 8886, 2021, doi: 10.1038/s41598-021-88257-w.
- [5] A. M. Alaa, T. Bolton, E. Di Angelantonio, J. H. F. Rudd, and M. van der Schaar, "Cardiovascular disease risk prediction using automated machine learning: A prospective study of 423,604 UK Biobank participants," *PLoS ONE*, vol. 14, no. 5, Art. no. e0213653, 2019, doi: 10.1371/journal.pone.0213653.
- [6] Y. Li, M. Sperrin, D. M. Ashcroft, and T. P. van Staa, "Consistency of variety of machine learning and statistical models in predicting clinical risks of individual patients: longitudinal cohort study using cardiovascular disease as exemplar," *BMJ*, vol. 371, Art. no. m3919, 2020, doi: 10.1136/bmj.m3919.
- [7] S. Barbieri, S. Mehta, B. Wu, C. Bharat, K. Poppe, L. Jorm, and R. Jackson, "Predicting cardiovascular risk from national administrative databases using a combined survival analysis and deep learning approach," *International Journal of Epidemiology*, vol. 51, no. 3, pp. 931–944, 2022, doi: 10.1093/ije/dyab258.
- [8] J. T. Neumann et al., "Cardiovascular risk prediction in healthy older people," *GeroScience*, vol. 44, pp. 403–413, 2022, doi: 10.1007/s11357021-00486-z.
- [9] J. S. Suri et al., "A powerful paradigm for cardiovascular risk stratification using multiclass, multi-label, and ensemble-based machine learning paradigms: A narrative review," *Diagnostics*, vol. 12, no. 3, Art. no. 722, 2022, doi: 10.3390/diagnostics12030722.
- [10] M. M. Islam, S. Kumar, M. A. Salam, D. C. Roy, and M. R. Karim, "Cardiovascular risk prediction and influencing predictors identification among Bangladeshi individuals using machine learning algorithms and association rule mining," *PLoS ONE*, vol. 20, no. 10, Art. no. e0333913, 2025, doi: 10.1371/journal.pone.0333913.
- [11] P. Shah, M. Shukla, N. H. Dholakia, and H. Gupta, "Predicting cardiovascular risk with hybrid ensemble learning and explainable AI," *Scientific Reports*, vol. 15, Art. no. 17927, 2025, doi: 10.1038/s41598025-01650-7.
- [12] S. Tasnim, M. Mamun, S. H. Chowdhury, M. I. Hussain, and M. M. Hossain, "Advancing interpretable AI for cardiovascular risk assessment: A stacking regression approach in clinical data from Bangladesh," *Medinformatics*, pp. 1–11, 2025, doi: 10.47852/bonviewMEDIN52027812.